Notes: Chari and Henry (JF, 2004)

Risk Sharing and Asset Prices: Evidence from a Natural Experiment

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1 Big picture

This paper studies whether systematic risk matters for asset prices by using stock market liberalizations as a natural experiment. When an emerging market opens to foreign investors, the relevant pricing portfolio for investible stocks changes from the local market portfolio to the world market portfolio. If international risk sharing lowers a firm's systematic risk, the firm's required return should fall and its stock price should rise.

The core empirical fact is that firms made eligible for foreign purchase, called *investible firms*, experience large stock-price revaluations around liberalization. The average investible firm rises by about 15.1% in the liberalization month, while the average noninvestible firm rises by about 9.9%. The paper asks whether the extra revaluation is explained by the firm-specific fall in systematic risk rather than only by a common country-level shock.

One-sentence summary. This note summarizes Chari and Henry (2004), which uses firm-level data from emerging-market stock liberalizations, an investible/noninvestible comparison, and the difference between local-market and world-market covariances to show that international risk sharing explains a substantial fraction of the stock-price revaluation of firms opened to foreign investors.

2 Research question

The paper asks three related questions:

1. When countries liberalize stock markets, do stock prices respond to the change in systematic risk predicted by international CAPM logic?
2. Are revaluations of investible firms larger when their covariance with the local market is much greater than their covariance with the world market?
3. Can the observed price effects be separated into a common shock, such as a fall in the risk-free rate, and a firm-specific risk-sharing effect?

The key insight is that aggregate studies observe only one revaluation per country, so they cannot cleanly distinguish common discount-rate changes from firm-specific risk-sharing changes. Firm-level data provide cross-sectional variation in exposure to risk-sharing gains.

3 Incremental contribution of the paper

This paper makes four main contributions.

1. **It shifts the asset-pricing test from levels to changes.** Standard cross-sectional asset-pricing tests ask whether average returns are related to beta levels. This paper instead asks whether stock prices change when the relevant source of systematic risk changes discretely.
2. **It uses stock market liberalization as a natural experiment.** Liberalization changes the marginal investor and the pricing benchmark from the domestic market to the world market for investible firms.
3. **It separates common and firm-specific revaluation channels.** The common component captures broad discount-rate changes, while the firm-specific component is measured using the covariance difference $DIFCOV$.

4. **It exploits the investible/noninvestible distinction.** Investible firms should be priced by foreign investors using world-market risk; noninvestible firms provide a comparison group that helps rule out pure market-wide price pressure.

Relative to earlier aggregate liberalization papers, the central advance is identification. The paper does not merely show that liberalizations are followed by higher stock prices; it asks whether the magnitude of each firm's revaluation is proportional to the predicted reduction in systematic risk.

4 Theoretical framework

4.1 Before liberalization

Before liberalization, the domestic equity market is segmented. Domestic investors price local stocks using the domestic market portfolio. The required return on firm i is

$$E[\tilde{R}_i] = r_f + \beta_{iM}\left(E[\tilde{R}_M] - r_f\right), \tag{1}$$

where β_{iM} is firm i's beta with the domestic market. With constant relative risk aversion, the local-market risk premium can be written as $\gamma\sigma_M^2$, giving

$$E[\tilde{R}_i] = r_f + \beta_{iM}\gamma\sigma_M^2. \tag{2}$$

4.2 After liberalization

After full integration, the relevant systematic risk is the world market portfolio. Firm i's required return becomes

$$E[\tilde{R}_i^*] = r_f^* + \beta_{iW}\gamma\sigma_W^2. \tag{3}$$

Subtracting the post-liberalization required return from the pre-liberalization required return gives

$$\Delta E[\tilde{R}_i] = E[\tilde{R}_i] - E[\tilde{R}_i^*] = (r_f - r_f^*) + \gamma DIFCOV_i, \tag{4}$$

where

$$DIFCOV_i = cov(R_i, R_M) - cov(R_i, R_W). \tag{5}$$

Interpretation. Liberalization affects required returns through two channels. The term $(r_f - r_f^*)$ is common to all firms in a country. The term $\gamma DIFCOV_i$ is firm-specific and measures how much the firm benefits from moving from local-market risk pricing to world-market risk pricing.

5 Main variables

- **Stock price response:** the abnormal return in the liberalization month, or cumulatively over the liberalization month and the following month.
- **Investible dummy:** equals one for firms that become legally and practically available to foreign investors.
- **DIFCOV:** historical covariance with the local market minus historical covariance with the world market.
- **DIFCOV1:** for noninvestible firms, covariance with the local market minus covariance with the portfolio of noninvestible firms.
- **Controls:** earnings surprise, firm size, liquidity/turnover, and country fixed effects.

6 Data and sample construction

The sample contains 410 firms from 11 emerging markets: Argentina, Brazil, Chile, Colombia, India, Korea, Mexico, Pakistan, Taiwan, Turkey, and Venezuela. The paper uses firm-level returns from the IFC Emerging Markets Data Base. Firms must have at least five years of pre-liberalization data so that historical covariances can be calculated.

Liberalization dates are country-level events after December 1988, because the IFC Investible Index begins at that time. Investible firms are those included in both the IFC Global Index and the IFC Investible Index, meaning they are legally available to foreign investors and satisfy practical size and liquidity screens.

7 Empirical design

The benchmark regression is

$$\Delta \ln(StockPrice_{ij}[0]) = \alpha + \beta_1 INVEST_{ij} + \gamma_1 DIFCOV_{ij} \tag{6}$$

$$+ \gamma_2 (DIFCOV \times INVEST)_{ij} + COUNTRY_j + \varepsilon_{ij}. \tag{7}$$

Reading the coefficients:

- α: common shock to all firms, such as a fall in the risk-free rate.
- $INVEST$: whether the common shock differs for investible firms. Theory predicts this should be insignificant.
- $DIFCOV$: risk-sharing effect for noninvestible firms.
- $DIFCOV \times INVEST$: additional risk-sharing effect for investible firms.
- $DIFCOV + DIFCOV \times INVEST$: total risk-sharing effect for investible firms.

The paper estimates specifications with country fixed effects and standard errors adjusted for clustering within country-liberalization events.

8 Main results

8.1 Average revaluation

Investible firms experience an average real dollar stock-price response of 15.1% in the liberalization month. Noninvestible firms experience an average response of 9.9%. This raw difference is consistent with the idea that investible firms benefit more from the shift to global risk sharing.

8.2 Risk-sharing channel

The average $DIFCOV$ for investible firms is about 0.018. In the preferred two-month specification with controls, the total effect of risk sharing for investible firms is about 3.59. Multiplying 0.018 by 3.59 gives approximately 0.065, or 6.5 percentage points. This is roughly two fifths of the 15% average revaluation.

8.3 Noninvestible firms

The risk-sharing coefficient for noninvestible firms is small and statistically insignificant in the main regressions. This is important because it shows that the covariance-based explanation works mainly for the firms actually opened to foreign investors.

8.4 Price pressure concern

If investible firms rise simply because they are included in indices and face foreign demand pressure, the investible dummy should capture an extra common shock for investible firms. The investible dummy is generally insignificant, which weakens the pure price-pressure explanation.

9 Interpretation and economic mechanism

The paper's mechanism is simple but powerful. A domestic investor in a segmented market cares about how a firm covaries with the local market. A foreign marginal investor in an integrated market cares about how the firm covaries with the world market. If a firm has high covariance with the local market but low covariance with the world market, its systematic risk falls sharply when foreign investors become marginal buyers. Its discount rate falls, and its stock price rises.

This logic explains why $DIFCOV$ predicts the revaluation of investible firms. The paper is not saying that every stock rises only because of risk sharing. A common shock also matters. The contribution is to show that the firm-specific part of the revaluation lines up with firm-specific risk-sharing fundamentals.

10 Extracted figures and tables from the paper

10.1 Table I: Decomposition of DIFCOV and Figure 1: Investible-firm repricing

Table I
Decomposition of DIFCOV

This table decomposes DIFCOV into its constituent parts. The variable "Number of Firms" is the number of firms in the specified country. All other numbers are the average value of the given variable across all of the firms in the specified country. All variables are computed at a monthly frequency. $\text{cov}(R_i, R_M)$ is the covariance of firm i with the local market. The variable $\text{cov}(R_i, R_W)$ is the covariance of firm i with the world market; ρ_{iM} is the correlation of firm i with the local market; ρ_{iW} is the correlation of firm i with the world market; σ_i is the standard deviation of firm i's stock return; σ_M is the standard deviation of the market; σ_W is the standard deviation of the world market.

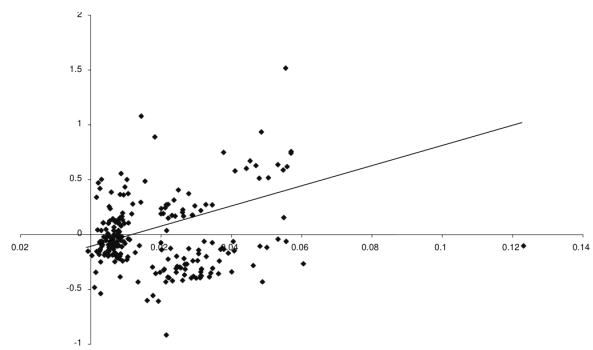
	Number of Firms	DIFCOV	$\text{Cov}(R_i, R_M)$	$\text{Cov}(R_i, R_W)$	ρ_{iM}	ρ_{iW}	σ_i	σ_M	σ_W
Panel A: Means, Investible Firms									
Sample	238	0.0183	0.0185	0.0001	0.6202	0.0602	0.1955	0.142	0.047
Argentina	14	0.0476	0.0473	−0.0003	0.9048	−0.0171	0.3486	0.295	0.045
Brazil	21	0.0283	0.0289	0.0005	−0.0924	0.0107	0.2906	0.205	0.044
Chile	9	0.0065	0.0070	0.0005	0.6787	0.0905	0.1252	0.079	0.047
Colombia	5	0.0025	0.0029	0.0004	0.6613	0.0210	0.1079	0.077	0.048
India	39	0.0097	0.0097	0.0000	0.6382	0.0017	0.1411	0.099	0.045
Korea	66	0.0059	0.0061	0.0002	0.5934	0.1326	0.1613	0.086	0.047
Mexico	4	0.0053	0.0055	0.0001	0.1587	−0.0093	0.1584	0.292	0.044
Pakistan	3	0.0002	0.0010	0.0008	0.5868	0.0127	0.0941	0.058	0.048
Taiwan	63	0.0281	0.0283	0.0002	0.8094	0.0854	0.2146	0.158	0.049
Turkey	9	0.0425	0.0414	−0.0010	0.7305	−0.0312	0.2503	0.205	0.053
Venezuela	5	0.0174	0.0177	0.0003	0.6893	−0.1003	0.2115	0.145	0.049
Panel B: Means, Noninvestible Firms									
Sample	172	0.0096	0.0105	0.0010	0.4250	0.0479	0.1921	0.142	0.047
Argentina	10	0.0382	0.0384	0.0002	0.7983	−0.0127	0.3683	0.295	0.045
Brazil	23	0.0214	0.0225	0.0011	−0.0963	0.0472	0.3061	0.205	0.044
Chile	17	0.0059	0.0067	0.0008	0.5459	0.1145	0.1366	0.079	0.047
Colombia	15	0.0024	0.0028	0.0004	0.5079	0.0464	0.1412	0.077	0.048
India	23	0.0126	0.0127	0.0001	0.6641	−0.0145	0.1752	0.099	0.045
Korea	6	0.0059	0.0064	0.0004	0.5877	0.0402	0.1878	0.086	0.047
Mexico	27	−0.0025	0.0014	0.0039	0.1025	0.3089	0.2375	0.292	0.044
Pakistan	38	0.0004	0.0007	0.0003	0.5196	−0.0701	0.0930	0.058	0.048
Taiwan	0	NA	NA	NA	NA	NA	NA	0.158	0.049
Turkey	5	0.0462	0.0456	−0.0006	0.7453	0.0127	0.2580	0.205	0.053
Venezuela	8	0.0162	0.0160	−0.0002	0.6746	−0.1259	0.1858	0.145	0.049

Equation (5) also predicts that the revaluation should be an increasing function of DIFCOV.[5]

Figure 1 reveals that the stock price revaluation for investible firms is an increasing function of DIFCOV, as theory predicts. The figure plots the unexpected stock price change for investible firms on the y-axis and DIFCOV on

[5] Since the coefficient of relative risk aversion is assumed to be the same across countries, the slope coefficient is also implicitly the same and therefore does not require a country-specific adjustment.

(a) Table I: Decomposition of DIFCOV



re 1. Differences between covariances with the market and world portfolio help ain the repricing of investible firms.

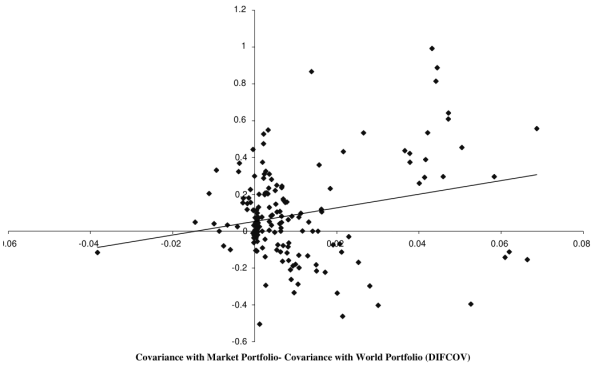
(b) Figure 1: DIFCOV and investible-firm repricing

Read:

- Table I decomposes $DIFCOV$ into covariance, correlation, and volatility components for investible and noninvestible firms.
- The average investible firm's local covariance is much larger than its world covariance, so liberalization offers large potential diversification gains.
- Figure 1 plots liberalization-month revaluation against $DIFCOV$ for investible firms and shows a clear positive slope.

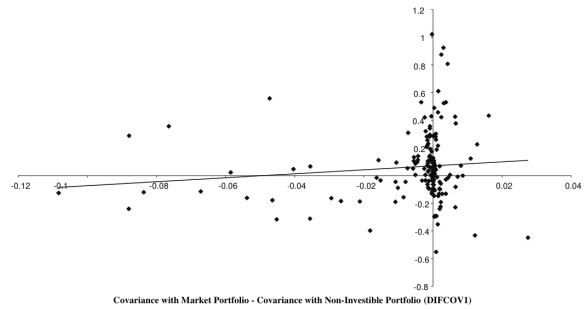
Economic message: Investible firms provide substantial global diversification benefits. The larger the gap between local and world covariance, the larger the predicted fall in required return and the larger the observed stock-price revaluation.

10.2 Figure 2: Noninvestible firms and Figure 3: DIFCOV1 for noninvestible securities



ure 2. Differences between covariances with the market and world portfolio do not p explain the repricing of noninvestible firms.

(a) Figure 2: DIFCOV and noninvestible-firm repricing



gure 3. Differences between covariances with the market and noninvestible portfolio not help explain the repricing of noninvestible securities.

Additionally, there is a more general concern. The goal is to estimate the npact of liberalization on a randomly selected firm from the population of all rms. If the investible firms are not randomly selected, then they may have un-

(b) Figure 3: DIFCOV1 and noninvestible-firm repricing

Read:

- Figure 2 shows a weaker relationship between $DIFCOV$ and revaluation for noninvestible firms.
- Figure 3 uses $DIFCOV1$, the covariance difference involving the noninvestible portfolio, and finds little explanatory power.

Economic message: The risk-sharing mechanism is much stronger for investible firms than noninvestible firms. This supports the view that the pricing change works through foreign investor eligibility rather than a generic market-wide repricing of all firms.

10.3 Table II: Stock market liberalization dates and Table III: Mean and median stock price response

Stock Market Liberalization Dates

Each date corresponds to the earliest stock market liberalization that occurs after December 1988, which is the month when the IFC introduced its IFCI index.

Country	Date of Stock Market Liberalization
Argentina	September 1989
Brazil	May 1991
Chile	October 1989
Colombia	December 1991
India	November 1992
Korea	January 1992
Mexico	May 1989
Pakistan	February 1991
Taiwan	January 1991
Turkey	August 1989
Venezuela	January 1990

(a) Table II: Stock market liberalization dates

Table III
The Mean and Median Stock Price Response of Investible Firms to Liberalization Is Larger Than That o the Noninvestible Firms

The mean liberalization return is the average stock price change in Month [0]. t-statistics are given in parentheses. The median liberalization retu is the median stock price change in Month [0]. Column 4 in Panels A and B reports the number of firms that experienced liberalization month retur that were below their own historical median return. P-values for sign tests for the median returns are shown in parentheses. The final column repor results from a T-test of the difference in mean returns across investible and noninvestible firms in Month [0].

	Panel A: Investible Firms				Panel B: Noninvestible Firms				T-test of Difference in Mea
	Number Firms	Mean Lib Return	Median Lib Return	Number Negative	Number Firms	Mean Lib Return	Median Lib Return	Number Negative	Investible Versus Noninvestible
Entire Sample	248	0.151	0.121	43 (0.00)	181	0.099	0.086	83 (0.149)	Yes*
Argentina	14	0.639 (8.466)	0.609	0 (0.5)	10	0.391 (5.07)	0.302	0 (0.000)	Yes**
Brazil	21	−0.052 (−1.029)	−0.048	10 (0.00)	23	−0.198 (−3.802)	−0.048	11 (0.5)	Yes*
Chile	9	0.048 (1.20)	0.071	4 (0.5)	17	0.141 (5.023)	0.177	8 (0.5)	Yes**
Colombia	5	0.354 (4.223)	0.419	2 (0.5)	15	0.287 (6.312)	0.310	7 (0.5)	No
India	39	0.208 (4.683)	0.172	7 (0.00)	23	−0.024 (−0.435)	−0.054	17 (0.994)	Yes***
Korea	66	0.551 (11.37)	0.584	7 (0.00)	6	0.069 (0.406)	−0.047	3 (0.656)	Yes***
Mexico	5	0.288 (5.48)	0.319	2 (0.5)	36	0.247 (8.813)	0.223	12 (0.033)	No
Pakistan	5	−0.239 (−2.154)	−0.136	1 (0.188)	38	−0.01 (−0.478)	0.004	19 (0.564)	Yes***
Taiwan	69	0.329 (11.899)	0.332	4 (0.00)	0	N/A	N/A	N/A	N/A
Turkey	10	0.654 (8.532)	0.577	4 (0.00)	5	0.509 (10.01)	0.577	2 (0.5)	Yes***
Venezuela	5	0.446 (4.223)	0.451	2 (0.5)	8	0.478 (4.723)	0.451	4 (0.637)	No

(*), (**), and (***) refer to 10%, 5%, and 1% levels of significance, respectively.

(b) Table III: Liberalization-month revaluations

Read:

- Table II lists the liberalization month for each country in the sample.

- Table III compares raw liberalization-month returns for investible and noninvestible firms.
- Investible firms have a mean revaluation of 15.1%, compared with 9.9% for noninvestible firms.

Economic message: The raw event-study evidence already suggests that foreign-investor eligibility matters. The subsequent regressions ask whether this difference can be linked to diversification fundamentals.

10.4 Table IV: Pre-liberalization characteristics and Table V: Earnings growth after liberalization

Table IV
parison of Investible and Noninvestible Firms before Stock Market Liberalization

market capitalization as a fraction of total market capitalization; Liquidity is annual turnover; Earnings Growth is the
rnings; Market to book is the ratio of the market value of equity to the book value of equity; DIFCOV is the difference in cov
d world markets. The final column reports results from a t-test of statistical difference of the means of the two sub-samp

Investible Firms					Noninvestible Firms					Significant Diff
Mean	Median	Min	Max	Std. Dev.	Mean	Median	Min	Max	Std. Dev.	(t-Test mea
0.015	0.006	0.0004	0.129	0.023	0.014	0.006	0.0002	0.135	0.019	No
2.66	0.43	0.003	26.9	5.11	0.249	0.083	0	2.35	0.364	Yes***
0.18	0.08	−4.20	5.80	1.02	0.10	0.05	−4.01	6.50	1.11	Yes**
3.30	1.72	−1.27	78.0	6.12	1.48	1.03	−3.30	7.75	1.50	Yes***
0.018	0.01	−0.001	0.123	0.017	0.01	0.004	−0.038	0.069	0.016	Yes***

the 5% and 1% levels of significance, respectively.

(a) Table IV: Investible and noninvestible firm characteristics

Table V
Average Annual Growth Rate of Real Earnings Per Share around Liberalization

[+1], [+2], and [+3] report growth rates of real earnings in the first, second, and third year following the liberalization. Country-fixed effects are included in all regressions but are not reported. The F-test reports results about the statistical significance of the difference in the mean growth rates for investible and noninvestible firms.

	Investible	Noninvestible	Significant Difference? (F-Test)
[+1]	−0.217***	−0.393***	Yes**
	(0.066)	(0.090)	
[+2]	−0.230***	−0.113	No
	(0.068)	(0.099)	
[+3]	−0.106	−0.014	No
	(0.068)	(0.101)	
Constant	0.027	−0.036	
	(0.024)	(0.035)	

(**) and (***) refer to the 5% and 1% levels of significance, respectively.

(b) Table V: Earnings growth after liberalization

Read:

- Table IV shows that investible and noninvestible firms differ in liquidity, earnings growth, market-to-book, and $DIFCOV$, although not much in size.
- Table V checks whether future earnings growth differs after liberalization.

Economic message: Selection is a real concern because investible firms are not identical to noninvestible firms. The paper therefore controls for earnings, size, liquidity, and country effects when estimating the risk-sharing channel.

10.5 Table VI: Benchmark regressions

Table VI

Diversification Drives the Stock Price Revaluations for Investible Firms But Not for Noninvestible Firms

The LHS side variable in Panel A is the abnormal return in the liberalization month. The LHS variable in Panel B is the cumulative abnormal return in the liberalization month plus one month. The symbol DIFCOV is the difference between the historical covariance of firm *i*'s stock return with the local market and its covariance with the Morgan Stanley Capital Market Index (MSCI) world stock market index. Country-fixed effects are included in all regressions but are not reported. INVEST is a dummy variable that takes on a value of one for investible firms, and zero for noninvestible firms. The variable SIZE is defined as the ratio of that firm's market capitalization to the total domestic market capitalization as of one year prior to the liberalization date. The variable TURNOVER is defined as the sum of the dollar value of all shares traded over a 12-month period before the liberalization (*t*-24 to *t*-12) divided by that firm's total market capitalization. Earnings Surprise [+1] is the deviation of the growth rate of earnings from its pre-liberalization mean in year [+1] following the liberalization. Robust standard errors adjusted for clustering within parentheses.

	Panel A: Month [0] Window					Panel B: Month [0, +1] Window				
	(1a)	(2a)	(3a)	(4a)	(5a)	(1b)	(2b)	(3b)	(4b)	(5b)
DIFCOV	0.331	0.267	0.235	0.335	0.313	−0.403	−0.210	−0.258	−0.180	−0.222
	(1.01)	(1.104)	(1.103)	(1.098)	(1.099)	(1.231)	(1.340)	(1.336)	(1.341)	(1.339)
DIFCOV*INVEST	2.41***	2.407**	2.384**	2.612**	2.590**	4.42***	3.747**	3.708**	3.865***	3.812**
	(0.897)	(1.196)	(1.195)	(1.192)	(1.194)	(1.09)	(1.452)	(1.447)	(1.456)	(1.454)
Earnings Surprise [+1]		0.011	0.011	0.016	0.016		0.025	0.025	0.029	0.028
		(0.013)	(0.013)	(0.013)	(0.013)		(0.016)	(0.016)	(0.016)	(0.016)
SIZE*INVEST			−0.791		−0.630			−1.183*		−1.081
			(0.578)		(0.587)			(0.700)		(0.715)
SIZE*NONINVEST			0.528		0.238			0.931		0.772
			(0.759)		(0.765)			(0.919)		(0.932)
TURNOVER*INVEST				−0.001	−0.002				0.001	−0.0003
				(0.003)	(0.003)				(0.004)	(0.004)
TURNOVER*NONINVEST				−0.104***	−0.094**				−0.072	−0.052
				(0.040)	(0.041)				(0.048)	(0.050)
Constant	0.059***	0.061***	0.049**	0.091***	0.082***	0.073***	0.063**	0.044	0.083***	0.061*
	(0.015)	(0.022)	(0.024)	(0.024)	(0.028)	(0.018)	(0.026)	(0.029)	(0.030)	(0.034)
INVEST	0.006	0.003	0.032	−0.030	−0.007		0.033	0.078*	0.010	0.056
	(0.029)	(0.030)	(0.034)	(0.032)	(0.038)	(0.036)	(0.036)	(0.042)	(0.039)	(0.047)
Adjusted *R*-Squared	0.176	0.172	0.117	0.196	0.155	0.16	0.137	0.101	0.155	0.115
Number Obs.	410	410	410	410	410	410	410	410	410	410

(*), (**), and (***) refer to 10%, 5%, and 1% levels of significance, respectively.

Table VI: Diversification drives the revaluation of investible firms

Read:

- The dependent variable is the abnormal return in the liberalization month or the cumulative abnormal return over the liberalization month and the following month.
- $DIFCOV$ is insignificant for noninvestible firms.
- $DIFCOV \times INVEST$ is positive and significant across specifications.
- The investible dummy is generally insignificant, meaning investible firms do not receive a separate common shock beyond the country-level shock.

Economic message: The central prediction of the theory is supported: firm-specific diversification gains explain revaluations for investible firms, not for noninvestible firms.

10.6 Table VII: Alternative abnormal returns using MSCI world return and Table VIII: World-market-model abnormal returns

Table VII

Diversification Drives the Stock Price Revaluations for Investible Firms But Not for Noninvestible Firms

Abnormal Returns: Alternative Measure 1

This result continues to hold when abnormal returns are calculated relative to the MSCI world return in the liberalization month. The LHS side variable in Panel A is the abnormal return in the liberalization month. The LHS variable in Panel B is the cumulative abnormal return in the liberalization month plus one month. The abnormal returns are calculated by subtracting the returns on the MSCI world index from the firm returns in: (a) liberalization month and (b) liberalization month plus one month. The symbol DIFCOV is the difference between the historical covariance of firm i's stock return with the local market and its covariance with the Morgan Stanley Capital Market Index (MSCI) world stock market index. Country-fixed effects are included in all regressions but are not reported. INVEST is a dummy variable that takes on a value of one for investible firms and zero for noninvestible firms. SIZE is defined as the ratio of that firm's market capitalization to the total domestic market capitalization a of one year prior to the liberalization date. TURNOVER is defined as the sum of the dollar value of all shares traded over a 12-month period befor the liberalization ($t = 24$ to $t = 12$) divided by that firm's total market capitalization. Earnings Surprise [+1] is the deviation of the growth rate o earnings from its pre-liberalization mean in year [+1] following the liberalization. Robust standard errors adjusted for clustering within parentheses

	Panel A: Month [0] Window					Panel B: Month [0, +1] Window				
	(1a)	(2a)	(3a)	(4a)	(5a)	(1b)	(2b)	(3b)	(4b)	(5b)
DIFCOV	0.400	0.297	0.267	0.366	0.344	0.122	−0.174	−0.220	−0.142	−0.183
	(1.061)	(1.076)	(1.075)	(1.070)	(1.071)	(1.313)	(1.329)	(1.326)	(1.330)	(1.328)
DIFCOV*INVEST	2.041*	2.145*	2.117*	2.352**	2.321**	3.111**	3.410**	3.368**	3.535**	3.478*
	(1.152)	(1.166)	(1.165)	(1.162)	(1.163)	(1.425)	(1.441)	(1.436)	(1.444)	(1.442)
Earnings Surprise [+1]		0.008	0.008	0.013	0.012		0.022	0.022	0.026	0.025
		(0.013)	(0.013)	(0.013)	(0.013)		(0.016)	(0.016)	(0.016)	(0.016)
SIZE*INVEST			−0.674		−0.513			−1.089		−0.981
			(0.563)		(0.572)			(0.695)		(0.709)
SIZE*NONINVEST			0.791		0.502			1.097		0.930
			(0.740)		(0.746)			(0.912)		(0.925)
TURNOVER*INVEST				−0.001	−0.001				0.000	0.000
				(0.003)	(0.003)				(0.004)	(0.004)
TURNOVER*NONINVEST				−0.105**	−0.094**				−0.073	−0.053
				(0.039)	(0.040)				(0.048)	(0.049)
Constant	0.057***	0.06***	0.045*	0.090***	0.078***	0.032	0.040	0.019	0.060**	0.037
	(0.021)	(0.021)	(0.024)	(0.024)	(0.027)	(0.025)	(0.026)	(0.029)	(0.030)	(0.034)
INVEST	0.016	0.014	0.043	−0.020	0.004	0.049	0.044	0.088**	0.019	0.065
	(0.029)	(0.029)	(0.033)	(0.031)	(0.037)	(0.035)	(0.036)	(0.041)	(0.039)	(0.046)
Adjusted RSQ	0.114	0.111	0.072	0.142	0.110	0.095	0.094	0.069	0.106	0.079
Number Obs.	410	410	410	410	410	410	410	410	410	410

(*), (**), and (***) refer to 10%, 5%, and 1% levels of significance, respectively.

(a) Table VII: Alternative abnormal return measure 1

Table VIII

Diversification Drives the Stock Price Revaluations for Investible Firms But Not for Noninvestible Firms

Abnormal Returns: Alternative Measure 2

This result continues to hold when abnormal returns are calculated using the world market model estimated in the pre-liberalization window. Th LHS side variable in Panel A is the abnormal return in the liberalization month. The LHS variable in Panel B is the cumulative abnormal returi in the liberalization month plus one month. The abnormal returns are estimated using a world market return model, which is estimated, in th pre-liberalization window using the MSCI world index. The pre-liberalization window consists of returns in the $t = -72$ to $t = 0$ window. The symbo DIFCOV is the difference between the historical covariance of firm i's stock return with the local market and its covariance with the Morgan Stanle Capital Market Index (MSCI) world stock market index. Country-fixed effects are included in all regressions but are not reported. INVEST is dummy variable that takes on a value of one for investible firms and zero for noninvestible firms. SIZE is defined as the ratio of that firm's marke capitalization to the total domestic market capitalization as of one year prior to the liberalization date. TURNOVER is defined as the sum of the dolla value of all shares traded over a 12-month period before the liberalization ($t = -24$ to $t = -12$) divided by that firm's total market capitalization Earnings Surprise [+1] is the deviation of the growth rate of earnings from its pre-liberalization mean in year [+1] following the liberalization. Robus standard errors adjusted for clustering within parentheses.

	Panel A: Month [0] Window					Panel B: Month [0, +1] Window				
	(1a)	(2a)	(3a)	(4a)	(5a)	(1b)	(2b)	(3b)	(4b)	(5b)
DIFCOV	0.016	0.297	0.267	0.366	0.344	0.115	−0.180	−0.226	−0.149	−0.191
	(0.029)	(1.076)	(1.075)	(1.070)	(1.071)	(1.308)	(1.325)	(1.321)	(1.326)	(1.323)
DIFCOV*INVEST	2.04*	2.145*	2.117*	2.352**	2.321**	3.187**	3.485**	3.441**	3.605**	3.544*
	(1.061)	(1.166)	(1.165)	(1.162)	(1.163)	(1.420)	(1.436)	(1.431)	(1.439)	(1.437)
Earnings Surprise [+1]		0.008	0.008	0.013	0.012		0.022	0.022	0.026	0.024
		(0.013)	(0.013)	(0.013)	(0.013)		(0.016)	(0.016)	(0.016)	(0.016)
SIZE*INVEST			−0.674		−0.513			−1.066		−0.964
			(0.563)		(0.572)			(0.692)		(0.707)
SIZE*NONINVEST			0.791		0.502			1.194		1.037
			(0.740)		(0.746)			(0.909)		(0.922)
TURNOVER*INVEST				−0.001	−0.001				0.000	−0.0002
				(0.003)	(0.003)				(0.004)	(0.004)
TURNOVER*NONINVEST				−0.105***	−0.094				−0.073	−0.052
				(0.039)	(0.040)				(0.048)	(0.049)
Constant	0.018	0.021	0.006	0.051**	0.038	0.002	0.010	−0.013	0.030	0.005
	(0.021)	(0.021)	(0.024)	(0.024)	(0.027)	0.025	(0.026)	(0.029)	(0.030)	(0.033)
INVEST	0.015	0.014	0.043	−0.020	0.004	0.049	0.044	0.089**	0.020	0.067
	(0.029)	(0.029)	(0.033)	(0.031)	(0.037)	0.035	(0.035)	(0.041)	(0.039)	(0.046)
Adjusted RSQ	0.103	0.105	0.078	0.127	0.106	0.077	0.084	0.065	0.098	0.075
Number Obs.	410	410	410	410	410	410	410	410	410	410

(*), (**), and (***) refer to 10%, 5%, and 1% levels of significance, respectively.

(b) Table VIII: Alternative abnormal return measure 2

Read:

- Table VII subtracts the MSCI world return from firm returns.
- Table VIII estimates abnormal returns using a pre-liberalization world market model.
- In both robustness checks, $DIFCOV \times INVEST$ remains positive and statistically significant.

Economic message: The main result is not driven by global market movements around liberalization. The risk-sharing channel survives alternative definitions of abnormal returns.

11 How to connect this paper to broader literature

This paper sits at the intersection of international finance, asset pricing, and corporate finance.

- Compared with aggregate liberalization studies, it provides firm-level evidence that stock-price revaluations are partly due to changes in systematic risk.
- Compared with standard beta-pricing tests, it offers a stronger empirical design by focusing on a discrete policy shock that changes the relevant pricing kernel.
- Compared with price-pressure stories, it uses noninvestible firms as a comparison group to show that simple foreign demand pressure cannot fully explain the results.
- Compared with investment-based interpretations of stock prices, it suggests that changes in discount rates may guide future physical capital allocation after liberalization.

12 Strengths and limitations

12.1 Strengths

- The natural experiment gives the paper a clean economic shock.
- The firm-level design separates common country-level effects from firm-specific risk-sharing effects.

- The investible/noninvestible distinction provides a useful comparison group.
- The paper directly links theory to a measurable variable, $DIFCOV$.

12.2 Limitations

- Liberalization may be anticipated, so month-0 returns may understate the total effect.
- Investible firms differ from noninvestible firms, so selection cannot be fully eliminated.
- Revaluations may reflect changing cash-flow expectations as well as discount-rate changes, although the paper controls for earnings surprises.
- The analysis focuses on stock prices, not directly on welfare or real investment responses.

13 Takeaways for research

For asset pricing, the key lesson is that systematic risk can matter strongly when the test is designed around a large and observable shift in the pricing environment. For international finance, the key lesson is that capital-market integration generates heterogeneous gains across firms, depending on how each firm's risk loads on local versus world markets. For corporate finance, the paper suggests that stock-price signals after liberalization may guide the reallocation of physical investment toward firms whose cost of capital falls the most.

14 Short summary

Chari and Henry (2004) show that when emerging markets liberalize, investible firms experience large stock-price revaluations and these revaluations are systematically related to the predicted reduction in systematic risk. The variable $DIFCOV$, defined as local covariance minus world covariance, captures the firm-specific risk-sharing gain. Investible firms with higher $DIFCOV$ rise more, while noninvestible firms do not show the same robust relation. About two fifths of the average investible-firm revaluation can be explained by risk sharing.

15 Conclusion

The paper provides firm-level evidence that international risk sharing affects asset prices. Liberalization changes the relevant pricing benchmark from the local market to the world market for investible firms. Firms whose covariance with the local market is high relative to their covariance with the world market experience larger price increases. The result supports the view that financial globalization can reduce firms' costs of capital through improved risk sharing, not merely through a common fall in the risk-free rate or index-induced price pressure.